

Project Design Document

1. Design Intent

- *What experience are you trying to create?- we are creating an altered sensory experience of a popular high intensity sport where fun doesn't demand sweating. Usually drones are used either to race, or help spectate sports, but for the first time it bridges a gap between Xbox style sports and physical sports and acts as the ball itself. It does all the great things that a ball does with none of the inconveniences.*
- *Who is it for?- It's for everyone who can use a phone or gesture with their hands. As a rather idle game, it may appeal to people who aren't sporty but still want to appeal to*
- *If solving a problem, what problem does it solve?- Its main objective is to look at both drones and sports from a different perspective, but as a no contact quasi digital sport, it does solve problems people might have with actions like striking a ball due to injury/ disability etc.*

2. Inspiration & References



3. User Journey

The user starts with the touch of a button, interacting with the drone and initiating the “game”.

The system responds by initiating idle flight, it stays on standby for further instructions.

Again when initiated through the app, the parabola begins. The drone “draws” a parabola in the air, and reverses along the same path if it detects a “hit” from the player it was heading towards. As in traditional volleyball, a point is gained when the ball touches the ground, here, the ultrasound is the condition for the same, failing which, the point goes to the other player. A set of 5 and a game of 3 rounds/sets of which the winner takes 2.

4. Success Criteria

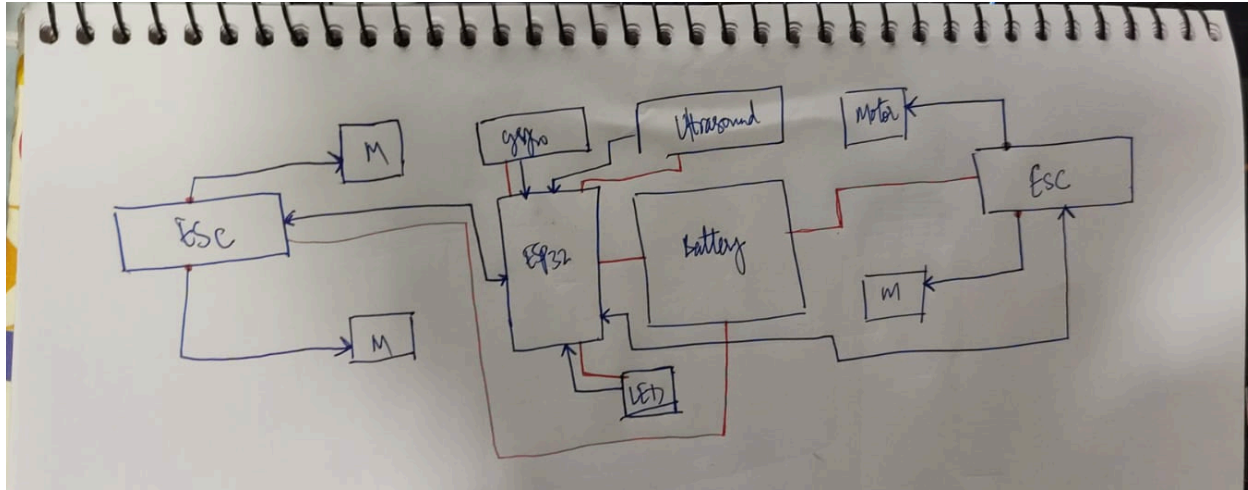
- smooth transitioning of flight states (under 5 seconds)
- simple architecture enabling easy hardware fixes/modifications
- ultrasound measures 3 varying forces (not mandatory)
- Easy-to-follow signifiers on the app w.r.t buttons and animations
- App stays relevant during gameplay: bold and attention-grabbing, but does not distract players (simple and minimalistic)
- Corrective strategies help stabilise the drone even in faster wind simulations (failsafe)

5. System Overview

This system is an assembly of a logical centre and its computational parts which control its hardware’s movement together. The ESP32 receives calculations and corrections from the accelerogyroscope and ultrasound and logically determines the next course of action as directed. The ESP32 manages simultaneous instructions to be given to the hardware and parallelly also calculates for if/then decisions

Major components of the technology apart from the esp32 itself are the escs and the accelerogyroscope. They work together to maintain both stable flight against wind and external air resistance.

6. High-Level Block Diagram



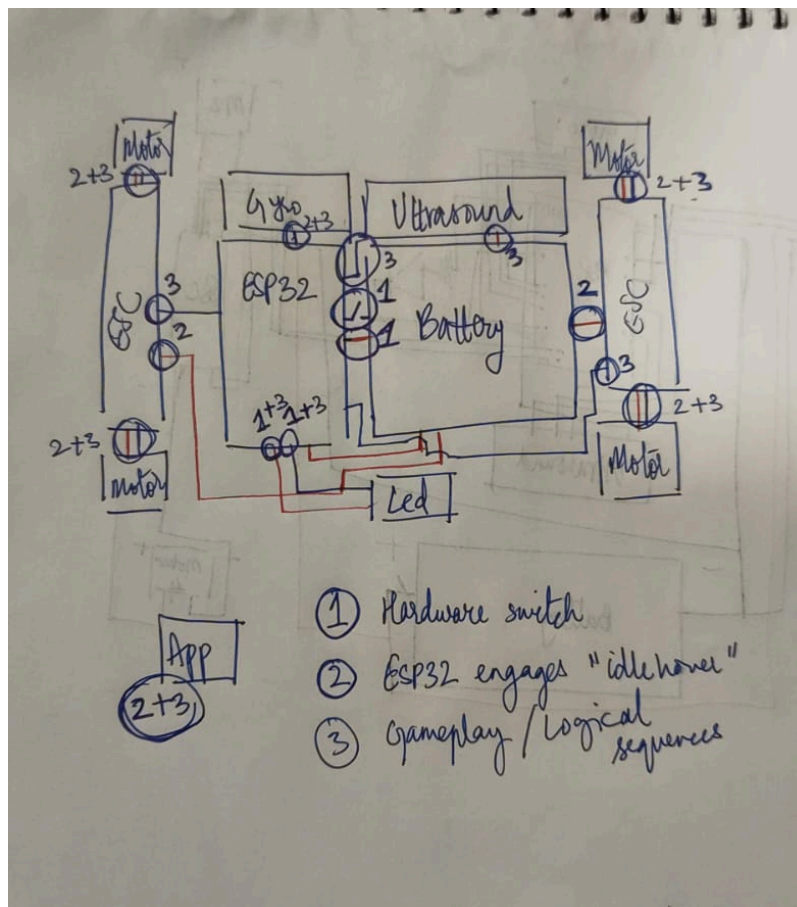
👉 Visual diagram of system components and flow

7. Input / Output Mapping

	Input	Process	Output
"Hit" magnitude	Ultrasound Distance between hand and sensor in 50 ms. Ultrasound Distance between hand and sensor in 30 ms.	The difference of these values is put into 4 different categories, 3 based on how fast the receiver has moved his hand (inversely related to difference), and 1 being the hit "fail"	Either the parabola changes direction to reverse based on 1/2/3, if category 4 is reached, the drone goes into timeout idle hover.
accelerogyroscop e	Roll, Pitch, Yaw values being calculated currently	Compared against "control values" of corresponding drone positions. error difference values are calculated	The position variables, if different from the control, are corrected using corresponding error difference values

parabola	Accerlogyroscope values	Here control values are calculated by being put in the parabola equation, controlling roll, pitch, yaw, and throttle.	The parabola now moves in a definitive "path" as opposed to the idle hover.
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9. Final System Architecture (Labelled)



Project Planning & Management Document

1. Project Overview

· *Project Title: Volleybee*

Medha: 3D fabrication, idle hover and timeout/shutdown code

Asmita: Parabola code, App

the project involves creating a drone with simulates a game of volleyball. It recognises “hits”/”defenses” and distributes scores accordingly,

2. Team Roles & Responsibilities

Member Name	Role	Responsibilities
Medha Nair	- Fabrication - Code	3D printing drone frame Idle Hover and Timeout logic, code
Asmita Wankhede	- Software - Code	MIT app inventor flow, code, design Parabola, bluetooth, ultrasound logic, code

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
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Week 0	Understand drone functions and logics + Component Requirements.	A comprehensive, basic sense of the physics in drone mechanisms and how that translates to computer data.	Complete
Week 1	Code Research + Development Frame Fabrication	Learning how to write equations as operative instructions for drone motors. Writing up code drafts of the same for various flight modes. Designing the drone Frame	Complete
Week 2	Hardware Fabrication and Circuit Setup + Testing	3D printing the drone frame	Complete
Week 3	Debugging		

4. Weekly Progress Log

Week 0

- *Planned Tasks:*
 1. *Research about drone mechanisms and working, required technologies*
- *Completed Tasks: 1*
- *Challenges Faced: there was an overload of information, and it was quite misleading on the whole because many technologies' implementation was done with arduinos and components we weren't already familiar with, and so we had to research further for alternatives.*
- *Solutions: We were able to get a hang of the thrust, the corrective strategies and required components for the drone (which we could afford)*
- *Next Steps: An optimum frame and a start on the logic and planned circuitry.*

Week 1

- *Planned Tasks:*
 1. *Build frame*
 2. *logic formulation: parabola and overall game logic and flow.: study the physics behind the movement and research about how it can be converted and applied to thrust and corrective values (accelerogyroscope)*
- *Completed Tasks: 1, 2*
- *Challenges Faced: Since this step involved incorporating actual theoretical concepts from physics into hardware movements, there were a lot of logical complications as to **what exactly needed to undergo the “conversion” of thrust as per the parabolic equation, because we also had to keep in mind that the corrective strategy would also be influenced by the same.***
- *Solutions: after quite a few logic iterations and research, and some help from ai as to how to frame the logic, we were able to come up with a few*
- *Next Steps: start with codes*

Week 2

- *Planned Tasks:*
 1. *Hardware: build final circuitry and start testing.*
 2. *Software: finish with basic logic codes separately (separate files)*
- *Completed Tasks: 2*
- *Challenges Faced: We didn't realise yet that we were missing a few components - the hardware seemed to be working, but it was extremely uncooperative. We couldn't really decipher what the esc beeps meant at that time. We had a bit of a knowledge backlog. The hardware worked separately, the parts just would not work together.*
- *Solutions: We researched about the components we needed and started building the app and debugging the code through “secondary” means: through other code references and AI.*
- *Next Steps: assemble the hardware and perform testing.*

Week 3

- *Planned Tasks: run code on the final circuitry, and assemble the parts on the drone.*
- *Completed Tasks: -*
- *Challenges Faced: this is where we realised we did not have enough power for all the components: the ESCs, the esp32 itself, and the accelerogyroscope. The wired connections*

were also weak because of the breadboard which we tried troubleshooting. We needed a larger battery and therefore better heat management system, maybe a separate heat sink, and/or a larger drone frame altogether.

- Solutions: we couldn't fix this in the remaining time and went ahead with experimenting with other, simpler ideas.
- Next Steps:

Test Plan & Results

1. Test Plan

Test	Method	Expected Result
<i>What are you testing? (Usability, performance, etc.)</i> Idle hover/flight	Test motors and ESCs and send a regular thrust signal after arming ESCs	The drone achieves a certain height (as we determine) . It does not necessarily need to work completely in sync with the corrective strategy but the motors and escs would need to work in perfect coordination. There cannot be any delays.
Failsafe : parabola to idle hover	Conduct the ultrasound test and record reaction success and time taken to deduce whether optimisation would be required to protect escs from sudden thrust shifts and battery from irregular power draws	The drone converts to this flight state in a reflex amount of speed, not just to imitate actual volleyball movement but also for the safety of the components themselves. A slower reaction time is also alright, more preferred, because a reflex-level speed involves a large power-draw change and sudden irregular movement.

2. User Testing

For as long as we had the setup working, it was just the two of us, we had to assume the role of users too.

We evaluated for weight and pressure sensitivity of the ultrasound. The app was also tested for user experience.

3. Results

Test	Outcome	Pass/Fail
Ultrasound	The ultrasound should be able to detect 2 different distances and final output to be out of 3 different “push intensities”: where the difference is directly proportional to “force” of the hit action	pass
Motors	defect checking, simultaneous working at specific speeds	pass, fail
ESC	basic connection speed control and modulation of motors	fail
Gyroscope/ Accelerometer	Drone stabilisation	fail
circuitry	Optimum flow of current so as to preserve battery and heat management	pass

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4. Observations

- *User behavior*
- *Pain points: flight states would not smoothly transition into each other, the escs would not deliver instructions fast enough, making flight jerky and unpredictable.*

6. Improvements Suggested

- *Larger base for components instead of deeper*
- *Better heat management*

Learnings & Reflections Document

1. Team Reflection

- *How did the team work together?*

Since we were working with newer technologies, and we had different skill backgrounds we relied on each other to collectively understand the drone working.

- *What worked well?*

We were able to assemble the circuitry fairly well enough on our own, and had the code almost figured out too. Our ability of logical deduction and trial-and-error helped.

- *What could be improved?*

We should have planned out and organised our research better (so that we don't miss out on crucial information again) and run simulation tests so we know exactly what we need instead of directly assembling/purchasing and then failing.

2. Technical Reflection

- *Key technical challenges*

Heat management and flow of current was a problem because of our drone's physical architecture and the number of components. We weren't able to properly optimise the flow of current and the parts got hot almost immediately.

- *What did you learn (hardware/software)?*

Optimising flow of current was a huge priority because our operation time is really long and we're working with a limited power source attached to quite demanding components.

- *Mistakes & debugging experiences*

We should've performed online simulation testing before finalising the components because they weren't optimum for our drone's size, the power, and the collective weight of the drone. The motors would have needed to spin much faster to accommodate it all otherwise, and we didn't have enough power for that, and it would generate an extremely dangerous amount of heat and be unsafe for handling too.

3. Design Reflection

- *Did the experience match your intention?*

For the most part, yes. It was small and cutesy and portable too, not intimidating newer users and staying true to its purpose of being for all ages and for leisurely gameplay rather than defense or security operations like its military counterparts. The bright orange allowed it to remain fairly distinct in its testing environment.

- *What would you improve?*

Give it a more unique design language and optimise it for the larger components - make it deep-seated but not with a huge surface area.

- *User feedback insights*

To make it easier to handle.

4. Personal Reflection

Asmita.

I was last a student of physics in the 10th grade, and most of this experience felt like “OH WAIT I REMEMBER THIS.” There were tens of smaller concepts that we solved individually instead of trying to figure out everything at once and the solutioning process became simpler than what we anticipated it to be.

The detail-heavy mechanism nature of a drone felt quite overwhelming, because we were learning something new at every step, and we knew virtually nothing about it before. Seeing concepts as they are and solving them on paper was easy enough, but the real struggle lay in how differently the same concept acted out in the real world, and how we needed to change how we saw the problem, rather than forcing a logic that we had formed beforehand.

We had to rely a lot on our own ability to code and understand but I think it was really cool of us to not shy away from attempting something nobody we knew had done before and that we believed in ourselves and our ambition till the very end.

Medha:

I learnt about the intricacies of what it takes to make something fly. From choosing the right power distribution system (ESC) to coding a hyperbolic path, the journey was filled with the truths of practical application as compared to theory. When creating a frame for the drone, I learnt how to bring components together in fabrication and the details that make it cohesive.

While there were many things pursued and corrected along the course, we realized there were also the logistical issues of testing and finding the right components, especially the battery with a suitable voltage. Fixing everything to theoretically be able to fly was only one step closer to actually making it fly, and our project managed to keep surprising us with challenges like weight, overheating and voltage distribution.

I am extremely proud of our attempt at making this drone and our sustained research into how we can improve on our work. I learnt so much about something I wouldn't have expected myself to even try at the start of this year, and it goes to show